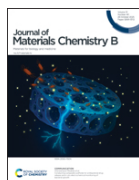


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A biocompatible, eco-friendly 3D-printed PCL-NIPUA resin for macular bucking devices in myopic tractional maculopathy



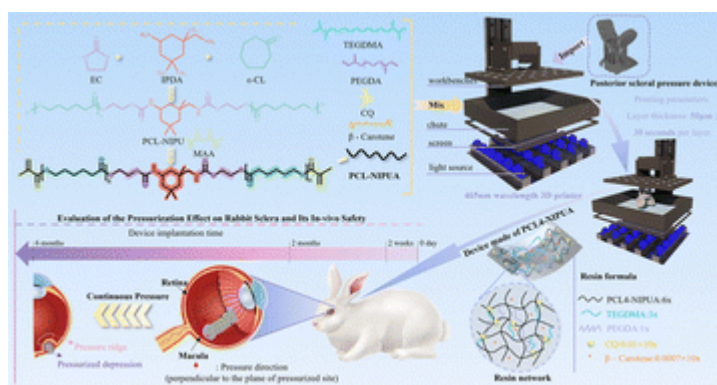
[Zi Fu](#),^{†,abc} [Yan Zhou](#),^{†,de} [Ziyi Liu](#),^{abc} [Zhe Chen](#),^{de} [Qiang Fu](#),^f [Shiqun Lin](#),^{de} [Rongping Dai](#)^{*de} and [Huade Zheng](#) ^{*abc}

[Author affiliations](#)

Abstract

Pathological myopia, characterized by excessive axial elongation, posterior staphyloma (PS) and retinal-choroidal degeneration, underlies sight-threatening conditions such as myopic traction maculopathy (MTM) and macular atrophy. Macular buckling (MB) surgery has demonstrated significant efficacy in managing MTM-associated pathologies, including macular schisis, macular holes, and macular retinal detachment. However, its widespread clinical adoption remains limited due to suboptimal material properties and geometrical design constraints of existing buckling devices. Advancements in imaging modalities – such as spectral-domain optical coherence tomography (SD-OCT), swept-source OCT (SS-OCT), ultra-widefield fundus imaging, and three-dimensional magnetic resonance imaging (3D-MRI) – have substantially enhanced the understanding and evaluation of MTM biomechanics. The future development of MB may lie in identifying superior biomaterials and creating customized buckling devices tailored to individual ocular morphology. Here, we developed a photo-curable polycaprolactone (PCL)-based non-isocyanate polyurethane acrylate (NIPUA) resin (PCL-NIPUA) for one-piece J-shaped 3D-printed MB devices. Structural characterization *via* nuclear magnetic resonance (NMR), Fourier transform infrared (FTIR) spectroscopy, and mass spectrometry (MS) confirmed successful synthesis. Photocuring kinetics under 465 nm visible light achieved an optimal storage modulus within 82 s, coupled with a minimal curing shrinkage (7.2%). By tuning flexible PCL

segments, the material exhibited a tensile strength of 25 MPa and an elongation at break of 32%, balancing strength and elasticity. The resin demonstrated environmental stability (water contact angle: 72°; water absorption: 8.8%; and thermal decomposition: 420 °C) and exceptional biocompatibility (hemolysis rate <0.5%; cell viability >97%). Liquid crystal display (LCD) 3D printing produced cost-effective MB devices that provided stable scleral indentation in rabbit models, with few complications, and no pathological structural changes in the optic nerve and retina, showing good biocompatibility and safety. This work establishes PCL-NIPUA as a synergistic platform combining green chemistry, rapid photocuring, mechanical adaptability, and biosafety, offering a clinically translatable solution for customized 3D-printed implants in MB surgery.



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